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AMSC N/A

REVIEWER ACTIVITIES
AIR FORCE-82
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ARMY-CR, MI

DD FORM 1 MAY 72 **1072-1** (CONDENSED)

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5310-1541

P.A. NAVY-AS | INTERNATIONAL INTEREST

Other Cust ARMY-AV AIR FORCE-11

PROCUREMENT SPECIFICATION MIL-N-25027

SUPERSEDES NAS 697

TITLE NUT, SELF LOCKING, PLATE, TWO LUG, REDUCED RIVET SPACING, LOW HEIGHT, CRES, 125KSI Ftu, 450°F, & 800°F

MILITARY STANDARD

MS21070

PAGE 1 OF 2

FED SUP CLASS 5310

Ⓕ **ENTIRE STANDARD REVISED**

CONFIGURATION OF NUT OPTIONAL WITHIN THE LIMITATIONS IMPOSED BY DIMENSIONS AND REQUIREMENTS SPECIFIED HEREIN

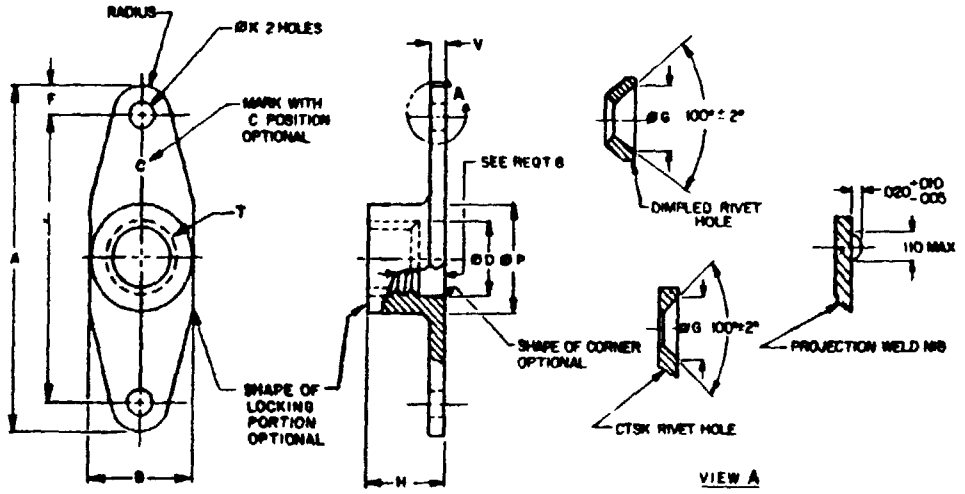


TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBERS						T THREAD SIZES	A MAX	B MAX	B MIN	Ø D MIN	F MIN	Ø G ± .010	H 1/ MAX	J 1/ ±.002	Ø R 1/ +.005 -.000	Ø P MAX	V MAX	AXIAL STRENGTH LBS/MIN	HEIGHT LBS/100
PLAIN HOLES	100° CSK OR DIMPLED HOLES	WELDING PROJECTION	PLAIN HOLES	82° CSK OR DIMPLED HOLES	WELDING PROJECTION														
L02	—	L02W	-02	—	-02W	.086-56 UNRC-3B	418	.162	.140	—	071	4/	.110	250	.066	.130	.035	440	.05
L03	—	L03W	-03	—	-03W	.099-48 UNRC-3B	425	.184	140	—	071	4/	125	265	066	.143	035	550	08
L04	L04K	L04W	-04	-04K	-04W	.112-40 UNRC-3B	.630	.260	.140	—	089	200	.143	406	.098	.186	.040	750	.11
L06	L06K	L06W	-06	-06K	-06W	.138-32 UNRC-3B	661	.265	.195	—	089	200	171	.437	.098	.206	.047	1,130	14
L08	L08K	L08W	-08	-08K	-08W	.164-32 UNRC-3B	692	.297	.230	168	100	200	.250	468	098	.248	.047	1,720	18
L3	L3K	L3W	-3	-3K	-3W	.190-32 UNRP-3B	724	.328	298	194	100	200	250	500	098	.274	.047	2,460	26
L4	L4K	L4W	-4	-4K	-4W	.250-28 UNRP-3B	706	414	380	.254	.100	200	281	562	.098	.344	.055	4,580	.48
L5	L5K	L5W	-5	-5K	-5W	.3125-24 UNRP-3B	1 006	.505	475	317	125	230	328	718	130	.417	.065	7,390	.77
L6	—	L6W	-6	—	-6W	.375-24 UNRP-3B	1 116	614	580	379	125	230	344	.838	130	505	075	11,450	1 06

1/ MINIMUM "H" NOT SPECIFIED, LIMITED ONLY BY STRENGTH REQUIREMENTS OF SPECIFICATION.
2/ CENTER OF THE TAPPED HOLE SHALL NOT DEVIATE IN ANY DIRECTION FROM THE CENTER OF THE PLAIN HOLE AS DETERMINED BY THE RIVET HOLES BY MORE THAN .005.
3/ DIMPLED RIVET HOLE TOLERANCE FOR Ø "R" IS +.015 -.000.
4/ SPACE WILL NOT PERMIT CONVERSION.

APPROVED 6 APR 64 REVISED Ⓕ 30 SEP 87

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REQUIREMENTS:

1. MATERIAL: CORROSION RESISTANT STEEL, TYPE A286, (UNS S66286) IN ACCORDANCE WITH AMS5525, AMS5732 OR AMS5737.
2. FINISH: ONLY 800°F NUTS: SILVER PLATE ALL THREADED SURFACES OF NUT ELEMENT IN ACCORDANCE WITH AMS 2410. THREADS SHALL SHOW COMPLETE COVERAGE. SILVER PLATING ON ALL OTHER SURFACES OPTIONAL. TARNISHING OR DISCOLORATION OF SILVER NOT CAUSE FOR REJECTION.
3. LUBRICANT: ONLY 450°F NUTS: DRY FILM LUBRICANT IN ACCORDANCE WITH MIL-M-25027.
4. DIMENSIONING AND TOLERANCING: DIMENSIONING AND TOLERANCING SHALL BE IN ACCORDANCE WITH ANSI Y14.5M.
5. HARDNESS: 49HRC, MAX.
6. THREADS: THREADS BEFORE LUBRICATION IN ACCORDANCE WITH MIL-G-8879.
7. SURFACE TEXTURE: SURFACE TEXTURE, UNLESS OTHERWISE SPECIFIED, SHALL NOT EXCEED 125 MICROINCHES, IN ACCORDANCE WITH ANSI B46.1.
8. COUNTERBORE/COUNTERSINK: ON SIZE .164 AND LARGER, THREAD RELIEF SHALL BE .062 MINIMUM; ON SIZE .138 AND SMALLER, COUNTERSINK TO "F" DIAMETER.
9. PERFORMANCE: SEE PROCUREMENT SPECIFICATION, EXCEPT 450°F DRY FILM LUBRICATED NUTS LIMITED TO FIVE REUSE CYCLES.
10. PART NUMBER: THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I.

EXAMPLE: MS21070L4W



MS21070L4W INDICATES: NUT, SELF-LOCKING, PLATE, TWO LUG, REDUCED RIVET SPACING, LOW HEIGHT, CRES; 125 KSI Ftu, 450°F DRY FILM LUBRICATION; .250-28 UNJF-3B; WELDING PROJECTION

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.
4. DESIGN AND USAGE LIMITATIONS: THESE NUTS ARE DESIGNED TO DEVELOP THE TENSILE STRENGTH OF BOLTS AND SCREWS WITH AN ULTIMATE TENSILE STRENGTH OF 125 KSI BASED ON THE CROSS SECTION AREA AT THE BASIC ROOT DIAMETER OF THE THREADS. THESE NUTS ARE DESIGNED TO BE USED ON 3A EXTERNAL THREADS. THESE NUTS SHALL BE USED IN ACCORDANCE WITH THE LIMITATIONS OF MS33588. ONLY NUTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL 25027 SHALL BE USED.

INTERCHANGEABILITY RELATIONSHIP

MS21070 NUTS CAN UNIVERSALLY REPLACE NAS697 NUTS OF LIKE MATERIAL, THREAD SIZE, LUBRICANT (DRY FILM OR NON-DRY FILM LUBRICANT), AND FASTENING METHOD (PLAIN RIVET HOLES; DIMPLED OR COUNTERSUNK RIVET HOLES), BUT THESE NAS697 NUTS CANNOT UNIVERSALLY REPLACE MS21070 NUTS.

INTERCHANGEABILITY TABLE

CANCELLED PART NUMBERS	SUBSTITUTIVE PART NUMBERS	CANCELLED PART NUMBERS	SUBSTITUTIVE PART NUMBERS
NAS697C02	MS21070-02	NAS697C6	MS21070-6
NAS697C02M	MS21070L02	NAS697C6M	MS21070L6
NAS697C03	MS21070-03	NAS697A3K	
NAS697C03M	MS21070L03	NAS697X4	
NAS697C04L	MS21070-04	NAS697X4K	
NAS697C04LM	MS21070L04	NAS697A4	
NAS697C06L	MS21070-06	NAS697A4K	
NAS697C06LM	MS21070L06	NAS697X5	
NAS697C08	MS21070-08	NAS697X5K	
NAS697C08M	MS21070L08	NAS697A5	
NAS697C3	MS21070-3	NAS697A5K	
NAS697C3M	MS21070L3	NAS697X6	
NAS697C4	MS21070-4		
NAS697C4M	MS21070L4		
NAS697C5	MS21070-5		
NAS697C5M	MS21070L5		

APPROVED 6 APR 64 REVISED (F) FOR CHANGES SEE PAGES 1 AND 2

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